Eval Report, AI Representative of Rajveer Bishnoi

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gpt-oss-120b (Groq) · bge-small embeddings (local) · FAISS · Cal.com booking

100%	93.2%	0.964	0	~1.2s
REFUSAL ACCURACY	GROUNDEDNESS	FAITHFULNESS	FABRICATED FACTS	TTFT MEDIAN

CHAT GROUNDEDNESS

58 questions across 8 areas: factual, why-hire, weakness, 30 project-detail (6 repos), 14 adversarial, and booking. Each answer is graded two ways: a deterministic fact-coverage check, and an **LLM judge that reads the real corpus** to verify every claim and classify refusals.

Refusal accuracy (traps / injection / out-of-scope)	100% · 14/14
Fabricated credentials (fake PhD / job / award)	0
Groundedness (answers fully corpus-supported)	93.2%
Mean faithfulness (claims supported)	0.964
Hallucination rate (LLM-judge)	6.9%
Retrieval hit-rate (top-k, search-path)	52%

On the hallucination number. A naive substring grader first reported 36-57%, all false positives: it flagged refusals that *name* a claim only to deny it ("no record of a PhD from Stanford"). The corpus-reading judge gives the true rate; ~half of the residual 6.9% are still conservative refusals, leaving roughly two minor over-claims and zero fabrications.

Retrieval. On the search-grounded subset, top-5 retrieval surfaced the expected repo 52% of the time. Factual questions bypass search via the deterministic fact tool, so they are excluded here.

VOICE QUALITY

First-response latency. On Groq, backend compute (LLM + embedding) is sub-second and streams, so TTS starts before the answer completes. The voice path is Vapi (US) to the Space (US), so network is negligible and first-response stays under the 2s target. A distant chat client adds ~0.9s of geographic India-to-US network (not compute).

Channels & transcription. Voice runs by phone and via a free in-browser call (same Vapi assistant, no dialing). STT: Deepgram `nova-3` (English).

Interruptions. Barge-in tuned (stop on 2+ words within 0.3s, 1.0s backoff, smart endpointing 0.4s), interruptions never crash the call.

Task completion. Booking (slots → confirm name/email/time → create) is verified end-to-end on the live Cal.com API and through the chat tool path; live-call sampling is the closing step.

THREE FAILURE MODES: FOUND & FIXED

1 · Technical embellishment under probing

Probed for internals it lacked (e.g. the private Samay Sarathi repo), the model invented specifics: "GCP + Docker + CI/CD", "Dijkstra/A* routing", a WebSocket schema. **Cause:** thin retrieval, so the LLM fills gaps. **Fix:** anti-embellishment + per-project separation + "internals are private" rules. **→ refusal 92.9% → 100%, faithfulness 0.884 → 0.964.**

2 · Confident false negatives

Asked "does he have teaching experience?", it said "no formal teaching roles", wrong; he is a Teaching Assistant. **Cause:** it answered from memory without checking the facts. **Fix:** a "check the tool before you deny" rule. **→ now leads with the TA / Buddy Program role.**

3 · Eval-grader false positives

The substring grader reported 36-57% hallucination, it can't tell "asserted X" from "denied X", and counted fact-path answers as retrieval misses. **Fix:** an LLM judge that reads the corpus, plus excluding fact answers from the retrieval denominator. **→ revealed the true 0%-fabrication, 100%-refusal picture.**

TRADEOFF MADE

All-free stack vs. paid low-latency. Everything runs on free tiers: Groq for sub-second gpt-oss-120b inference, a single-region HF Space for hosting, local embeddings, FAISS. The conscious tradeoff: free inference carries rate limits and a single US region adds geographic network latency for distant chat clients, accepted instead of paying for dedicated or multi-region capacity. Mitigated by retry + backoff, a semantic cache (paraphrases return in milliseconds), and the HF router as a fallback provider. Net: **$0 to run, fast where it matters** (the voice path), honest about the chat-from-afar network floor.

WITH TWO MORE WEEKS

- **Hybrid retrieval + re-ranking** (BM25 + dense + cross-encoder) to lift the 52% hit-rate and multi-fact recall, the biggest groundedness lever left.
- **Automated voice-eval harness**, scripted calls measuring WER, end-to-end latency, barge-in timing, and booking-completion rate.
- **Per-claim inline citations** (which repo, which line) for verifiable grounding.
- **Edge / multi-region deploy** to cut the ~0.9s network floor for geographically distant clients.
- **Richer corpus** built from commit history, not just READMEs.

Measured on the live deployment · 58-question golden set · LLM-judge grading over the real corpus · full code and test harness in the public repository.